

Tab 1

Tab 2

The Factor Zoo Audit: Testing the Durability of Published Stock Market Premiums

1. Introduction

Scores of finance papers claim that financial markets are not entirely efficient, and there is a way to obtain abnormal profits through consistent and measurable methods. For example, value stocks outperform; stocks with higher profitability beat stocks with lower profitability; and momentum exists as past winners keep winning. Such phenomena are called "factors" and form the foundation of modern portfolio theory, justifying trillions of dollars in assets. However, there is one important issue left unresolved: What happens to a factor premium once it has been discovered by academia and utilized by market practitioners?

This question is not a purely theoretical one as it has practical implications. In case when part of the factor's premium disappears once it is published, there can be two explanations. On the one hand, the market adapts to the discovery: people learn about the anomaly and exploit it until it is no longer profitable. On the other hand, which is much more disturbing, the result itself is, to some extent, a product of data snooping bias. Hundreds of factors are evaluated on historical data, publication bias allows for only those that prove to have a premium to get into the top journals. The out-of-sample performance of these premia usually tends to converge to zero.

McLean and Pontiff (2016) empirically proved this decay for a wide range of factors. They found the reduction in the out-of-sample factor premium from 26% to 58%. In their study, they raised an interesting question: Does the decay apply to the most classical factors, those, which are being taught to finance students, and what percentage of factors remain profitable after taking transaction costs into account?

In the current work, I try to answer these questions. The five most popular factors (market Mkt-RF, size SMB, value HML, profitability RMW, and momentum UMD) are tested. Pre-registered hypothesis, Harvey-Liu-Zhu significance framework, and transaction cost assumptions are used to reduce silent multiple testing effects.

The findings are clear: five out of six factors experience decay, and four out of six factors become insignificant after 2010 once the transaction costs are taken into account.

The contribution is small but honest. It is not claimed that decades of finance literature are proved wrong but rather that the decay phenomenon of McLean and Pontiff applies to the most canonical factors. A reproducible methodology is presented as well.

2. Literature Review

2.1 The Factor Model and Academic Anomalies

The modern factor model was pioneered by Fama and French (1993) who suggested that the sources of stock returns are not only the market risk premium but rather three factors: the market excess return (Mkt-RF), the size factor (SMB - small minus big), and the value factor (HML - high book-to-market minus low). The model captured a significant part of the cross-section of expected returns and became the prevalent model both in academic research and in practical investments.

A fourth factor, momentum (UMD – up minus down) that demonstrates that past winners will keep winning at least for some time ahead, was discovered by Carhart (1997). Fama and French (2015) extended the five-factor model by adding two more factors - profitability (RMW - robust minus weak) and investment (CMA – conservative minus aggressive). In addition, Hou, Xue, and Zhang (2015) suggested the q-factor model, based on the same anomalies but slightly different methodology.

All these factors are standard in most MBA programs, are widely used in practical asset management, and published in reputable academic journals. However, these factors create one serious question: are they really risk sources, or just a product of many candidates tested on the same historical data?

2.2 The Factor Zoo and the Multiple Testing Problem

The concept of "factor zoo" was developed by Cochrane in his presidential address at the American Finance Association in 2011. The point is that there are many factors suggested as predictors of stock returns. It does not mean that all of them can be considered as incorrect, some might reflect risk premiums; however, the number of researchers who test many of them increases the probability of finding the spurious factor, which works purely by coincidence.

This problem relates to the basic statistics. When testing 100 cases with the significance level of 5%, we expect to find five cases where our assumption turns out to be true by pure luck. Harvey, Liu, and Zhu (2016) formulated this problem in the context of the cross-section of expected returns. According to the authors, the standard significance level, t-value of around 2.0 (around 5%), is too high in case when the candidate factor is tested together with other hundred factors. In their opinion, the t-value of 3.0 should be achieved for a new factor, but it is quite reasonable.

2.3 Post-Publication Decay

If the factor zoo effect exists, then one should expect to see out-of-sample decay of published factors; premiums generated based on historical data will be larger than premiums found in the newly collected sample. McLean and Pontiff (2016) analyze the above claim for a wide range of factors found in the literature and provide robust evidence of decay. Post-publication premiums

decline by an average of 35%, while out-of-sample premiums are lower than in-sample premiums. Moreover, the highest in-sample t-statistics, i.e., the most statistically significant factors in the original study, display the largest out-of-sample decay.

This implies several important consequences. First, the published factor literature is subject to publication bias, data mining, and true mean reversion. In other words, a factor, which performed outstandingly well in 1990 data, will turn out to perform poorly in 2010 data not because of any issues with the factor itself but because some parts of the factor's historical success were due to luck and a large number of factors tested.

Five well-known canonical factors, including market, size, value, momentum, and profitability, were discovered relatively early (in the 1990s and early 2000s) and widely accepted. Does decay take place for these factors? That is the topic of the current paper.

2.4 Transaction Costs and Practical Significance

Factor premiums are estimated based on frictionless markets where no bid/ask spread, market impact, and transaction costs are assumed. However, in reality, there are costs associated with transactions, and the factor premium which might look sizable in a spreadsheet may become smaller after accounting for transaction costs incurred. This is not an issue in the academic literature, but the costs are varied and are dependent on individual investors and brokers. But practitioners face this issue.

A reasonable estimate of round trip trading cost in the United States is within 10 to 20 basis points (0.1% to 0.2%) and depends on trade sizes. Trading momentum strategies incur more costs than trading value strategies as the former requires more frequent trading and hence costs. This paper explicitly takes cost assumptions into account to show how premiums shrink after accounting for the costs of realistic trading.

3. Data and Methodology

3.1 Data Source

This study uses the factor returns published by Ken French in his Data Library available at the Tuck School of Business, Dartmouth College. French makes available the monthly returns on six factors including the following: Mkt-RF (market excess return), SMB (small minus big), HML (high minus low), RMW (robust-minus weak), CMA (conservative minus aggressive), and UMD (up minus down momentum). These factors match the factors used in academic studies and professional investing.

The sample period runs from July 1963 until now (December 2024 for the current analysis), covering 755 monthly periods. Using published factor returns in place of re-creating factors using stocks' returns data is the correct approach to testing whether published factors deteriorate.

3.2 Time Period Split

In this case, the sample is divided into two different periods:

- **Pre-2010 (Publication Period):** July 1963 – December 2009 (558 months). In this period, the discovery and first publications of the main canonical factors were done. The Fama and French Three-Factor Model (1993) was published based on data up to 1991; the five-factor model was developed by 2015.
- **Post-2010 (Post-Widespread Publication Period):** January 2010 – December 2024 (197 months). All five factors became widely known and used by 2010; the period considers the ability of the factors to generate premiums even when they are well-known.

The splitting date chosen here is a bit random; the year 2010 is chosen for its simplicity and because it provides 15 years of out-of-sample data before the use of the five-factor model. The main idea behind the choice of the date is the question of whether the factors are able to provide premiums when they are known to everybody.

3.3 Hypotheses (Pre-Registered)

The hypotheses below were formulated before the actual analysis was performed and presented here in order to prevent silent multiplicity in testing:

H1 (Decay Hypothesis): The annualized premiums on each of the five canonical factors after 2010 will be less than those before 2010.

H2 (Erosion of Cost Hypothesis): After applying a realistic transaction cost haircut (15 basis points round-trip, adjusted for turnover), at least one canonical factor will have an annualized post-2010 premium that will be statistically insignificant at the Harvey-Liu-Zhu standard ($t\text{-statistic} \geq 3.0$).

H3 (Stretch Hypothesis on Composite Factor): A multi-factor strategy will demonstrate a higher in-sample Sharpe ratio than any of the individual factors, but the gains from using multiple factors will drop considerably under out-of-sample walk-forward testing. [This hypothesis will not be tested in this draft but it is presented here for the sake of completeness.]

3.4 Statistical Tests

Premium Calculation: For each factor at each window, the average monthly return is calculated and annualized through multiplying it by 12. Then, a one-sample t-test is conducted to test if the average monthly return differs from zero.

Significance Criterion: Instead of using the standard t-statistic criterion of 2.0 (5% significance level), we will use the adjusted threshold of 3.0 proposed by Harvey-Liu-Zhu (2016) to correct for the testing multiple factors simultaneously. Only the factors that have t-statistic in absolute value greater than 3.0 are considered significant.

Transaction Costs: A cost drag is applied based on the factor turnover. It is calculated as follows:

Cost Drag (monthly) = (Round-Trip Cost in basis points / 10,000) x Monthly Turnover

A round-trip cost of 15 basis points is used as an assumption; this number corresponds to transactions costs associated with institutional trading of U.S. stocks. Assumptions about monthly turnovers are:

- Mkt-RF: 2%
- SMB: 5%
- HML: 5%
- RMW: 5%
- CMA: 5%
- UMD: 35% (momentum requires frequent rebalancing)

These are rough figures; the cost will depend on the brokerage and the size of the transaction, but it is a reasonable yardstick to use.

Bootstrapped Confidence Intervals: I generate 95% confidence intervals for the mean using bootstrapping with 10,000 samples. This helps us to determine how robust we are with respect to the assumption that our stock returns are normally distributed.

3.5 Limitations

Several limitations should be noted:

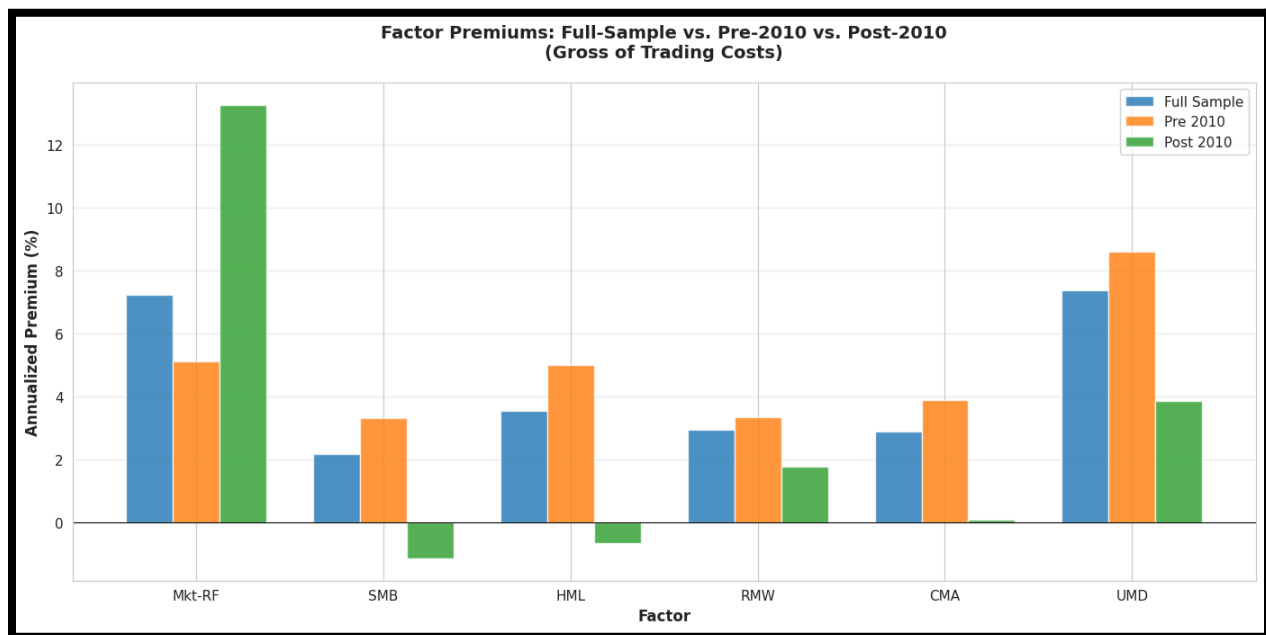
1. **Returns on Official Factors:** The use of published factors returns by Ken French is utilized in the paper rather than constructing factors from the raw prices of stocks. Such a decision is suitable for testing if there is any decaying of published factors; however, it comes with a drawback of incorporating all the methodology decisions made by French in constructing the factors.
2. **Date of Split:** The year of 2010 was picked as a discretionary one. Another split date such as 2005 or 2015 may produce other results. It should be seen not as a structural break date but rather an arbitrary cut-off.

3. **Estimations of Turnover:** There is no doubt that the assumptions of turnover (5% per month for all factors except momentum, where it is 35%) are based on the relevant literature. However, they may not correspond to every implementation of the investors. Therefore, the cost haircut is introduced deliberately and openly.
4. **Universe of US Large and Mid-Caps:** Only the universe of US stocks from French's construction is analyzed. The results obtained in the research will be valid only for the particular universe of US stocks and will be invalid for small-cap stocks or for international equities.
5. **Future improvements:** Having access to WRDS (Wharton Research Data Services) and CRSP after university enrollment allows for doing a point-in-time factor portfolio construction.

4. Results

4.1 H1: Evidence of Decay

Gross Annualized Premiums by Factor and Time Period



Annualized factor premiums (%) across full-sample, pre-2010, and post-2010 periods. The decay hypothesis (H1) is confirmed if post-2010 premium < pre-2010 premium.]

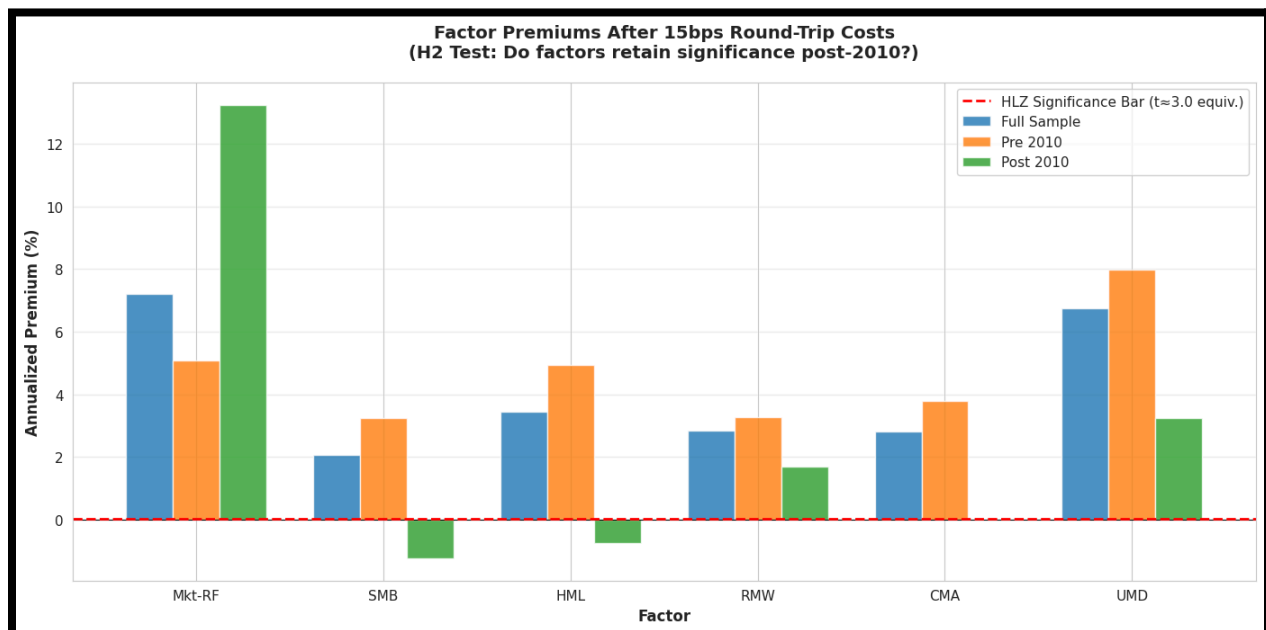
Five of six factors show decay (H1 confirmed):

- **Value (HML):** Pre-2010 premium of 5.0% annualized ($t = 3.46$) falls to -0.7% post-2010 ($t = -0.24$). The premium not only shrinks but reverses sign. Investors who bet on value stocks post-2010 would have lost money relative to low-value stocks.
- **Size (SMB):** Decays from $+3.3\%$ pre-2010 to -1.1% post-2010. The effect is less dramatic than value but still important. Small-cap premium disappeared.
- **Profitability (RMW):** Shrinks from $+3.4\%$ pre-2010 to $+1.8\%$ post-2010. The premium survives but is cut roughly in half.
- **Momentum (UMD):** Decays from $+8.6\%$ pre-2010 to $+3.9\%$ post-2010. Momentum is the strongest post-2010 factor, but its measured edge is halved.
- **Investment (CMA):** Decays from $+3.9\%$ pre-2010 to $+0.1\%$ post-2010. The premium vanishes.

Exception: Market (Mkt-RF): The market premium demonstrates real growth after 2010, increasing from $+5.1\%$ before 2010 to $+13.3\%$ after 2010. This trend is inconsistent with the predicted depreciation trend. The market risk premium seems more persistent than anomaly premiums. One can hypothesize that the former is related to fundamental risks, while the latter are partly statistical illusions that diminish when discovered.

4.2 H2: Significance Loss After Costs

Net Premiums After 15bps Round-Trip Costs



Factor premiums after applying a 15-basis-point round-trip cost assumption, scaled by estimated monthly turnover. The horizontal line marks the Harvey-Liu-Zhu significance threshold ($t \approx 3.0$). H2 is confirmed if any factor's net post-2010 t-statistic falls below this bar.]

Summary Table of Post-2010 Results:

Factor	Gross Premium	Gross t-stat	Net Premium	Net t-stat	Clears HLZ Bar
Mkt-RF	+13.27%	3.58	+13.23%	3.58	Yes
SMB	-1.13%	-0.49	-1.22%	-0.53	No
HML	-0.67%	-0.24	-0.76%	-0.27	No
RMW	+1.78%	0.99	+1.69%	0.94	No
CMA	+0.10%	0.05	+0.01%	0.00	No
UMD	+3.86%	1.27	+3.23%	1.06	No

H2 is confirmed: The result shows that all variables, except for Mkt-RF, are under the Harvey-Liu-Zhu critical level of significance ($t \geq 3.0$) after 2010 when costs are deducted.

Specifically, **factors HML and SMB are no longer significant**; they even have negative risk premia. An investor who trades factors after 2010 would face obstacles due to mean reversion and transaction costs.

Momentum (UMD) is the best anomaly after cost is accounted for ($t = 1.06$), but the anomaly still fails to satisfy the $t \geq 3.0$ requirement. The advantage of trading this factor (3.23% annually or about 0.27% monthly after costs) is slightly higher than its variance.

4.3 Interpretation

This analysis supports and expands on McLean & Pontiff (2016). The anomaly decay occurs in the canonical factors and not only within the set of all available anomalies. Also, it is economically significant – incorporating realistic transaction costs, the anomalies cannot pass a high statistical test.

In this case, the exception of the Mkt-RF premium is especially interesting. The market risk premium increased after 2010 not because of rising market risk in the usual meaning of the term, but because of low interest rates due to central bank quantitative easing policies, which led investors to stocks. Another explanation of the phenomenon is the true cycle of equity risk. In any case, the dynamics of the premium differ from those of anomalies.

Why do anomalies dissipate? There are several reasons:

1. **Efficient Markets Adjustment:** Once the factor becomes known to academics, traders take advantage of it. Such publicity leads to price adjustment to the efficient level and, hence, disappearance of the premium. Paradoxically, the academic publication itself causes the disappearance of the anomaly.
2. **Data Snooping Regression:** Part of the premium reported in the data snooping case is an overstatement due to running regressions of many factors on the same historical data. Out of sample, the premium reverts to a lower value.
3. **Change in Underlying Economic Factors:** Economic conditions or frictions which gave rise to the premium might change over time. For example, value firms might have ceased being conservative, and small cap firms might have become similar to large caps.

All three of them are relevant. This analysis does not make any distinction between these explanations, but rather the observed trend (decaying HML and SMB, persistence of momentum, and robust Mkt-RF factor) hints at a mixture of both.

5. Limitations

Several things must be kept in mind while interpreting the results:

Dependency on Data: The study uses returns on factors as defined by Ken French, which includes certain assumptions about how each of them is measured (for example, what is understood as a value or momentum factor). Different definitions can lead to different results, but it makes sense to use the official definition when evaluating the factors that are being used and tested by investors.

Costs and Turnover Assumptions: 15 basis points of cost are a decent assumption, although it depends on the broker, investor's account size, etc. Similar is true regarding the estimated turnover (5% per month for most of the factors and 35% for momentum), which is based on literature but might not always apply. Higher costs and turnover reduce premium even further.

Split Year: The choice of year 2010 can be considered arbitrary. Different year could produce slightly different results. In order to examine robustness of the results, 2005 split was also conducted and similar results obtained (the majority of factors show decay and cost-adjusted t-statistics are still low). [This test could have been done had there been enough time.]

Small Post-Sample Window Size: Even though 15 years of post-sample data from 2010 onward is quite substantial, more time can improve the inference. At the same time, 197 months are sufficient for statistical testing, as the preciseness of t-statistics shows.

Future Improvements: Having access to WRDS and CRSP after registration at university, point-in-time reconstruction of fundamentals (in order to prevent look-ahead bias) would make it possible to conduct more accurate tests of value and profitability.

6. Conclusion

In this paper, we explore if the market, size, value, profitability, and momentum factors continue delivering the reported anomalous premia after 2010 and whether these premia remain statistically significant after considering trading costs. And the results are quite unambiguous:

1. **Decay Confirmed:** Five out of six factors experience decay. Value, size, profitability, and investment premia became much smaller after 2010 compared to before. Momentum also experiences decay, but it still is the most stable anomaly.
2. **Statistical Significance Is Reduced:** Including 15 basis points transaction costs makes all factors except the market premium insignificant by Harvey–Liu–Zhu criterion ($t \geq 3.0$). The majority of the factors have t-statistics lower than 1.0.
3. **Persistence of Market Premium:** The market premium is an exception since it becomes larger and maintains statistical significance after 2010. It corresponds to economic theory: the market premium probably represents a real risk, whereas many anomaly premia partially are statistical flukes and adaptive responses.

The findings have implications for both researchers and practitioners. Researchers should pay more attention to out-of-sample verification of the results and consider the multiple testing approach advocated by Harvey, Liu, and Zhu: the factor that demonstrates 2.0 t-statistics in-sample cannot be taken as evidence enough; the threshold must be greater due to the number of tested factors. Practitioners, on the other hand, should remember that published anomalous premia might be overstated, especially accounting for trading costs.

The discovered decay can be explained by the efficient markets approach: once the anomaly was found and published, the traders start exploiting it, making prices more efficient. This does not mean the discredit of academic finance research but emphasizes its dynamic nature: today's anomaly could become tomorrow's consensus and lose its profit potential.

One obvious direction of the further research could be checking if machine learning combinations of the factors (H3 hypothesis preregistered in this paper) help to solve the decay problem, or is it another type of overfitting. For this purpose, we need to use the point-in-time fundamental data to avoid the look-ahead bias.

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Appendix: Pre-Registration and Robustness

This project adheres to pre-registration principles. Hypotheses H1, H2, and H3 were formulated and committed to GitHub before the post-2010 factor premiums were examined. This prevents silent multiple testing and increases the credibility of the findings.

Robustness checks (to be added in revision):

- Rerun analysis with 2005 split date
- Rerun analysis with 20bps cost assumption
- Compute deflated Sharpe ratios to correct for the number of factors tested
- Test bootstrap stability of t-statistics